

Open Spectrum In The Real World: The WiFi Explosion

Arguments about the benefits of open spectrum have in the past been largely theoretical. Techniques such as spread-spectrum were widely employed, but primarily in licensed bands or in military applications. Academic research showed the benefits of a spectrum commons. Without mass-market commercial examples, though, few were convinced the idea could fly in the real world.

Such real-world validation arrived in the form of WiFi and related wireless local area network (LAN) technologies. WiFi is a marketing and certification term promulgated by the Wireless Ethernet Compatibility Alliance, an industry trade group. It refers to the 802.11b and 802.11a wireless Ethernet standards defined by the Institute for Electrical and Electronic Engineers (IEEE). 802.11b, which was the first to take off commercially, operates in the 2.4 GHz Industrial, Scientific and Medical (ISM) band and delivers data speeds up to 11 megabits per second. 802.11a operates in the 5 GHz U-NII band and offers connections up to 54 megabits per second. Standards work in this area is ongoing, with proposed standards including 802.11g, which delivers higher-speed connections in the 2.4 GHz band, and 802.11e, which adds quality-of-service mechanisms to support high-quality voice and video delivery. The IEEE issued the final 802.11b standard in September 1999. The first mass-market commercial implementation, Apple's AirPort technology, went on the market that year. Since then, the market has grown rapidly, with expected sales of some 10 million PC/laptop adapter cards this year. Vendors such as Cisco, Linksys, D-Link, Netgear and Proxim are doing a brisk business selling access points for home networks, adding value to residential broadband connections. On the enterprise side, wireless LAN deployments doubled last year, with more than one million access points now in use in 700,000 companies, according to the Yankee Group. Cahners In-Stat sees the WiFi hardware market generating over \$5 billion in 2005, and that doesn't even include service revenues.